



ECEN 314 – Signals and Systems

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Module 15 - Application: Analog Filter Design
Part 1 – Butterworth LPFs

Learning Objectives



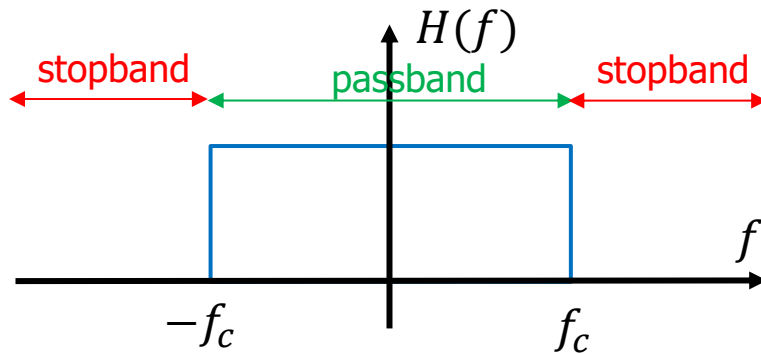
- Understand the form and characteristics of a Butterworth low-pass magnitude response.
- Given specifications on the passband and the stop band, be able to design a Butterworth LPF to meet those specifications.
- Given an n th order Butterworth low-pass filter design, be able to design a circuit to implement that response.

Filters

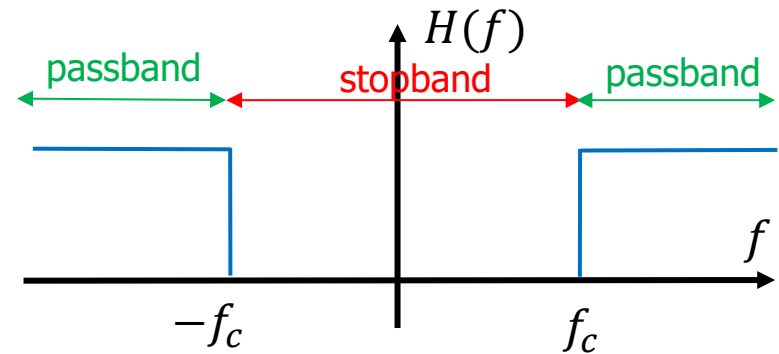


It is common to design circuits/systems to act as filters which pass certain frequency ranges and block (attenuate) others.

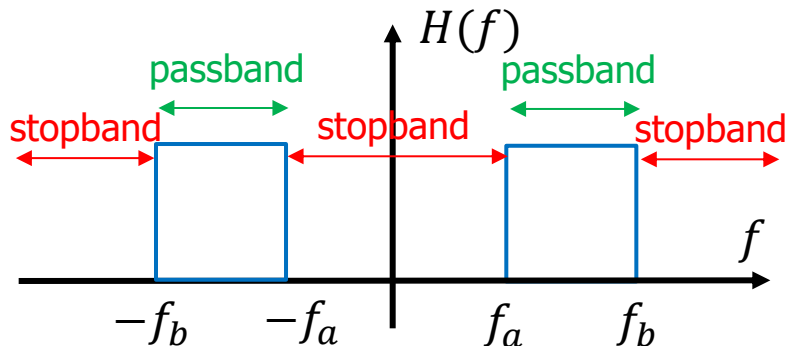
Lowpass Filter



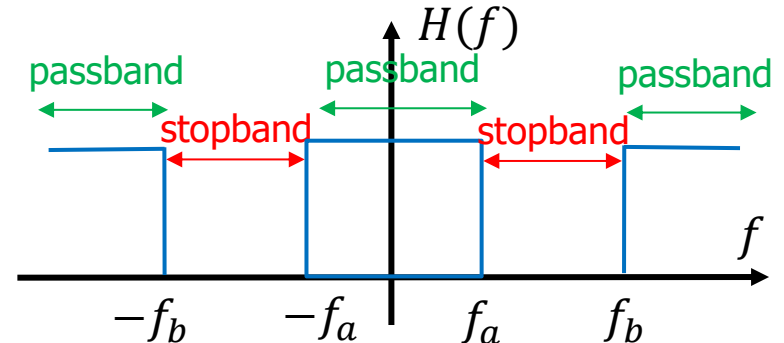
Highpass Filter



Bandpass Filter



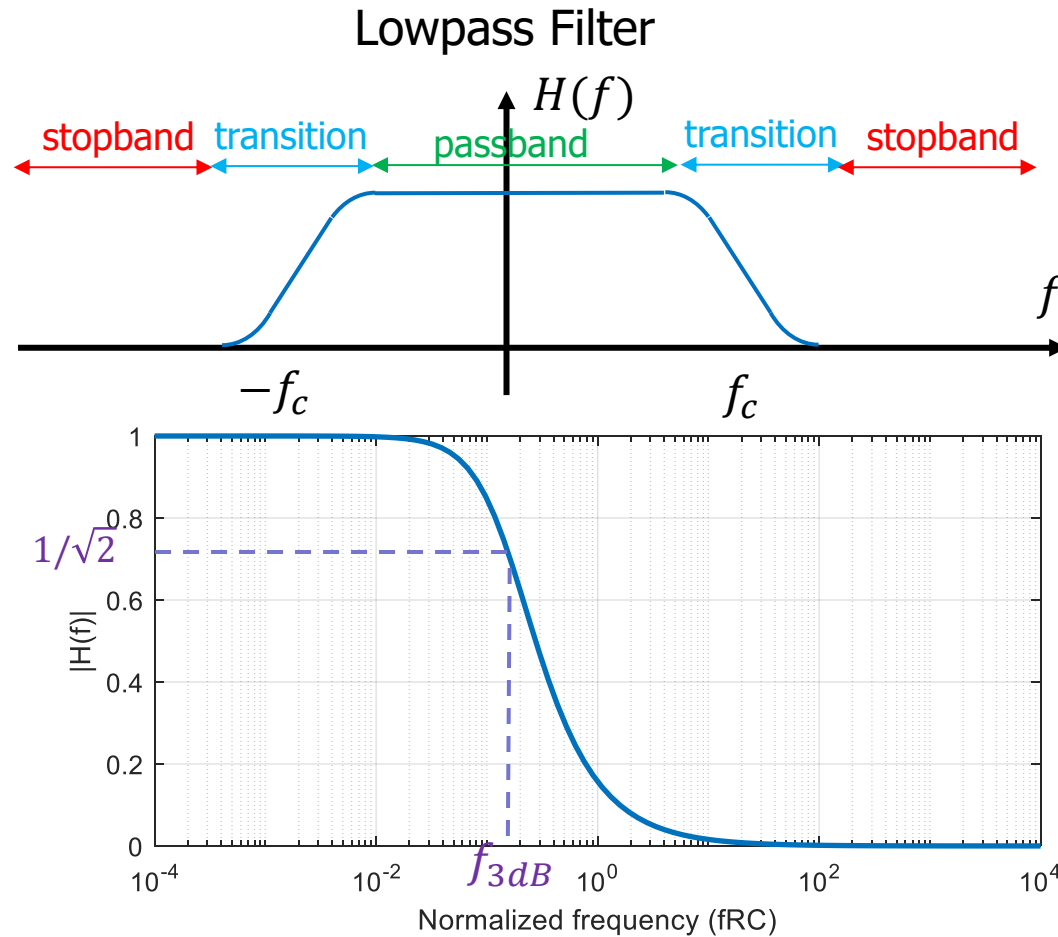
Band Stop (Notch) Filter



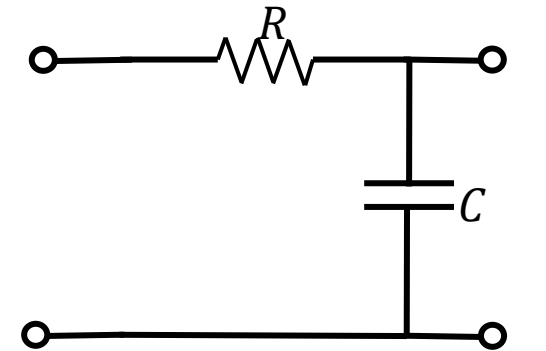
Filters



The ideal characteristics of the previous slide would be impossible to implement in practice, so we try to design filters that will function as a “good enough” approximation.



RC Lowpass Filter



$$H(s) = \frac{\frac{1}{sC}}{R + \frac{1}{sC}} = \frac{1}{1 + sRC}$$

$$H(f) = \frac{1}{1 + j2\pi fRC}$$

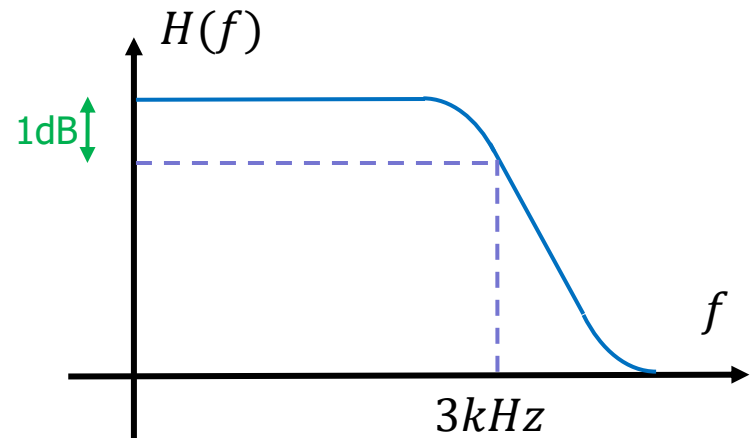
$$|H(f)| = \frac{1}{\sqrt{1 + (2\pi fRC)^2}}$$

RC Lowpass Filter Example



Example 15.1: Suppose we use a first order RC circuit as an LPF. We want the passband to be $|f| \leq 3kHz$ and we desire to have no more than 1dB attenuation throughout the passband.

- What should the 3dB frequency of the RC-LPF be?
- How much attenuation is achieved at $f = 20kHz$?



Answers:

- $f_{3dB} = 5.896kHz$
- $|H(20kHz)| = -10.97dB$

RC Lowpass Filter Example



Example 15.2: Suppose we want to design an LPF with the following specifications:

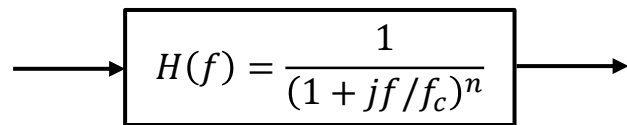
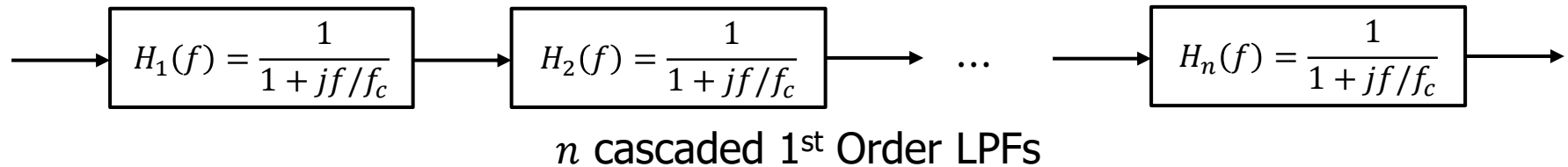
- ✓ Passband: $|f| < 10kHz$, with no more than 2dB attenuation throughout.
- ✓ Stopband: $|f| > 100kHz$, with at least 30dB attenuation throughout.

Can this be achieved with an RC lowpass filter?

Filters



In order to achieve the specifications of the previous example, it is necessary to use a higher order transfer function (e.g., a 2nd order circuit/system). One simple way to create an n^{th} order LPF would be to cascade n identical first order LPFs.



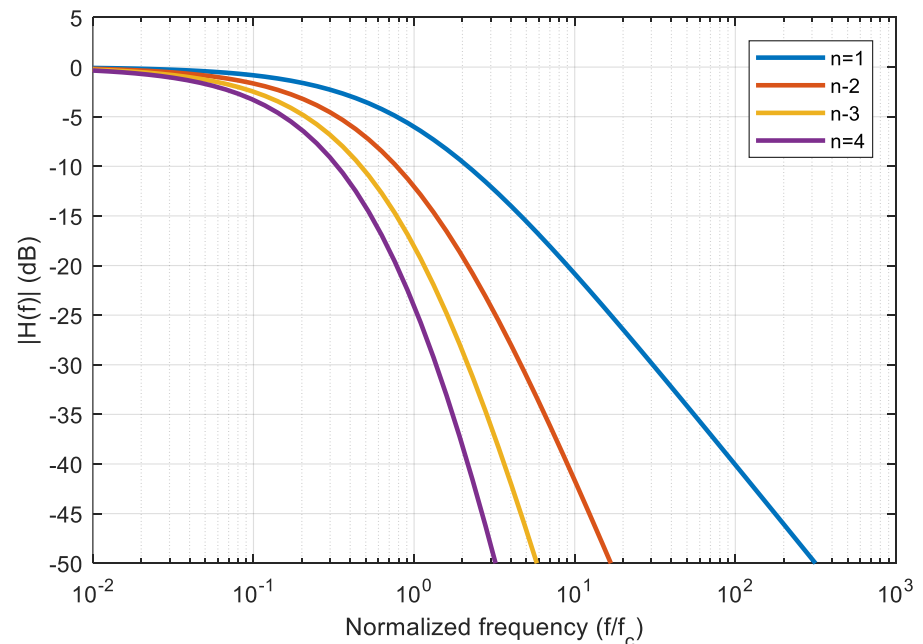
Equivalent n^{th} Order LPF

$$H(f) = \frac{1}{(1 + jf/f_c)^n}$$

$$\rightarrow |H(f)| = \frac{1}{(1 + (f/f_c)^2)^{n/2}}$$

or in the s-domain

$$H(s) = \left(\frac{-p}{s-p}\right)^n \text{ where } p = -\omega_c = -2\pi f_c.$$



Second Order Lowpass Filters



Another option for constructing a better higher order filter would be to include some complex poles. Suppose we built a second order filter with two poles at

$$p_1 = -\omega_c e^{j\theta} \text{ and } p_2 = -\omega_c e^{-j\theta}.$$

$$H(s) = \frac{-p_1}{s - p_1} \cdot \frac{-p_2}{s - p_2} = \frac{\omega_c^2}{(s + \omega_c e^{j\theta})(s + \omega_c e^{-j\theta})} = \frac{\omega_c^2}{s^2 + 2s\omega_c \cos(\theta) + \omega_c^2}$$

$$H(\omega) = \frac{\omega_c^2}{\omega_c^2 - \omega^2 + 2j\omega\omega_c \cos(\theta)} = \frac{1}{1 - \left(\frac{\omega}{\omega_c}\right)^2 + 2j\left(\frac{\omega}{\omega_c}\right)\cos(\theta)}$$

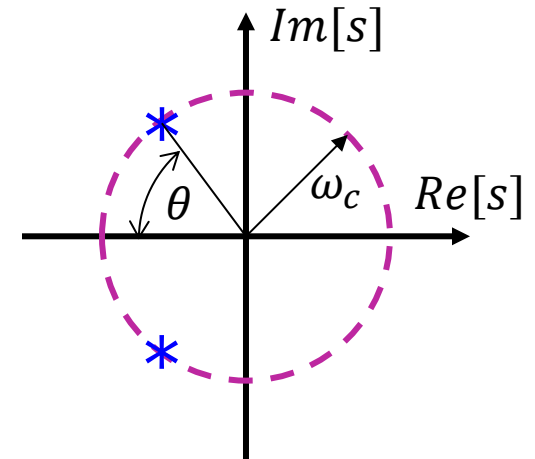
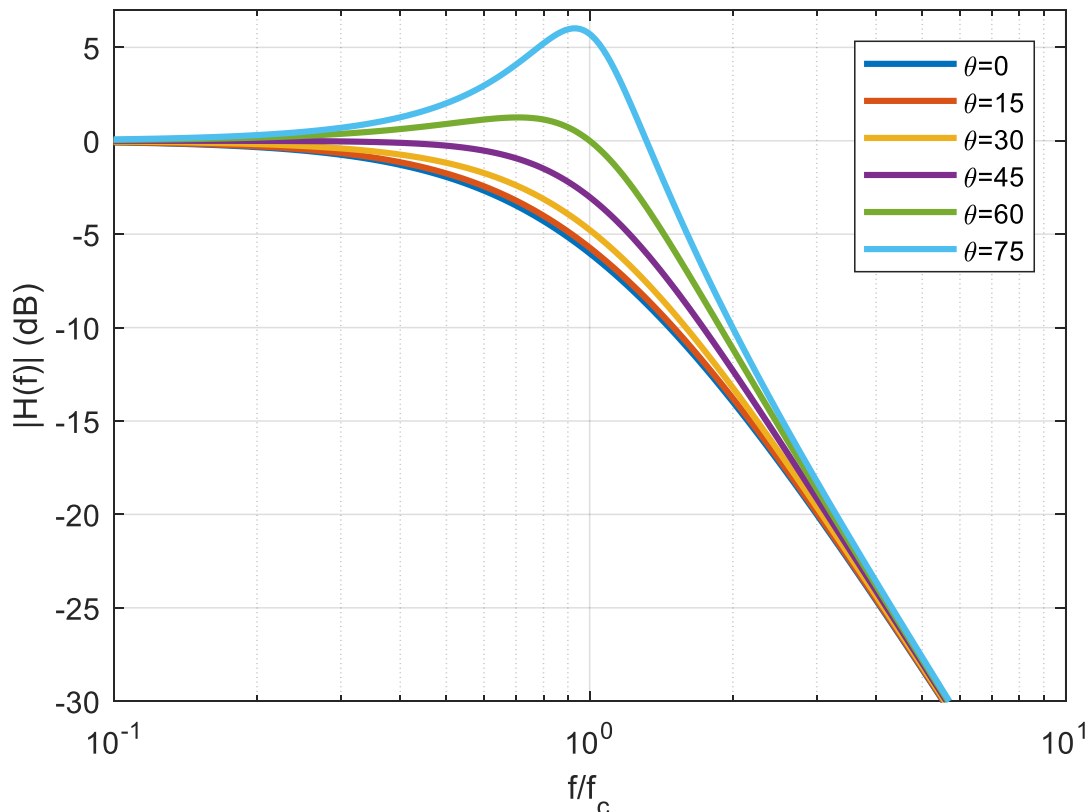
$$H(f) = \frac{1}{1 - \left(\frac{f}{f_c}\right)^2 + 2j\left(\frac{f}{f_c}\right)\cos(\theta)}$$

$$|H(f)|^2 = \frac{1}{\left[1 - \left(\frac{f}{f_c}\right)^2\right]^2 + \left[2\left(\frac{f}{f_c}\right)\cos(\theta)\right]^2}$$

Second Order Lowpass Filters



The plot below shows the magnitude response of a second order LPF with poles at $p = -\omega_c e^{\pm j\theta}$ for different values of θ (in degrees).



- ✓ Stopband behavior is the same for all θ .
- ✓ As θ increases, the passband gets wider (transition region gets narrower).
- ✓ If θ gets too large, there is too much fluctuation in the passband.

Butterworth Filters



There is a class of filters known as *Butterworth Filters* which are designed to have a “maximally flat” characteristic in the pass band. By maximally flat, it is meant that

$$|H(0)|^2 = 1,$$

$$\left. \frac{d^m}{df^m} |H(f)|^2 \right|_{f=0} = 0, \text{ for } m = 1, 2, 3, \dots$$

(for as many values of m as possible).

We seek to find the form of the transfer function ($H(s)$ or $H(f)$) that leads to this kind of magnitude response. Ultimately, we also want to know the pole (and zero) locations.

To start with,

$$H(s) = \frac{b_0}{b_0 + b_1s + b_2s^2 + \dots + b_ns^n} \rightarrow |H(f)|^2 = \frac{B_0}{B_0 + B_1f^2 + B_2f^4 + \dots + B_nf^{2n}}.$$

Next, if $|H(f)|^2$ is maximally flat at $f = 0$, then $|H(f)|^{-2}$ will also be maximally flat at $f = 0$.

$$|H(f)|^{-2} = \frac{1}{B_0} (B_0 + B_1f^2 + B_2f^4 + \dots + B_nf^{2n}).$$

Butterworth Filters



$$|H(f)|^{-2} = \frac{1}{B_0} (B_0 + B_1 f^2 + B_2 f^4 + \cdots + B_n f^{2n}).$$

It is not too hard to see that for this expression

$$|H(f)|^{-2} \Big|_{f=0} = 1, \quad \text{and} \quad \frac{d^m}{df^m} |H(f)|^{-2} \Big|_{f=0} = \begin{cases} 0, & m \text{ odd,} \\ \frac{m! B_m}{B_0}, & m \text{ even.} \end{cases}$$

Therefore, to make $|H(f)|^{-2}$ maximally flat (and thereby make $|H(f)|^2$ maximally flat), we should select $B_m = 0$ for $m = 1, 2, 3, \dots, n - 1$.

For an n^{th} order Butterworth LPF, the magnitude response is of the form

$$|H(f)|^2 = \frac{B_0}{B_0 + B_n f^{2n}} = \frac{1}{1 + (B_n/B_0) f^{2n}}.$$

We can express the constants B_n/B_0 in terms of the cutoff (3dB) frequency so that

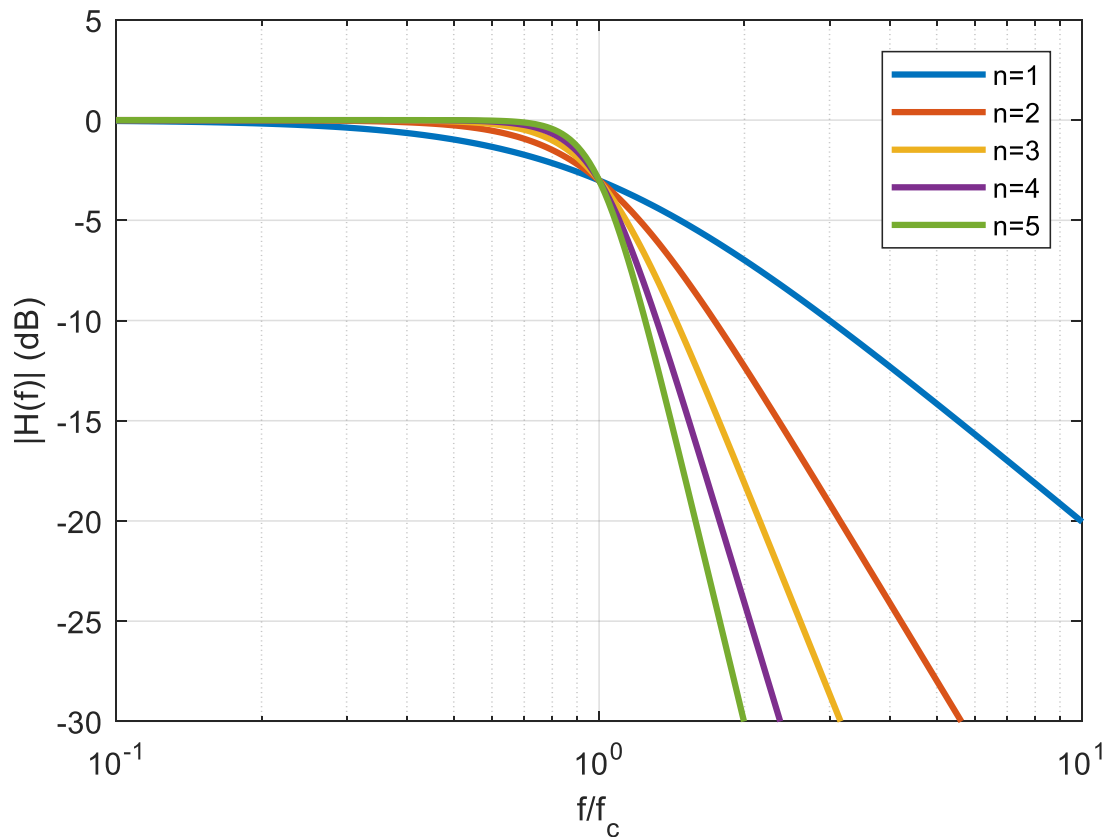
$$|H(f)|^2 = \frac{1}{1 + (f/f_c)^{2n}}.$$

Butterworth Magnitude Response



Butterworth Magnitude Response

$$|H(f)|^2 = \frac{1}{1 + (f/f_c)^{2n}}.$$



Characteristics of the Butterworth LPF

- ✓ Maximally flat response in the passband.
- ✓ All responses are 3dB down at $f = f_c$.
- ✓ $20*n$ dB/decade rolloff for $f > f_c$.
- ✓ As n increases, the response starts to look more and more like the ideal "brick wall" response.

Butterworth Pole Locations



Next, we seek to find a transfer function that leads to the desired magnitude response, $|H(f)|^2 = \frac{1}{1+(f/f_c)^{2n}}$.

Note that since $|H(f)|^2 = H(f)H^*(f)$, if $H(f)$ has a pole at some frequency $f = f_1$, then $|H(f)|^2$ will have a pole at the same frequency as well as another pole at $f = f_1^*$.

We can easily locate the poles of $|H(f)|^2$ by setting $1 + (f/f_c)^{2n} = 0$

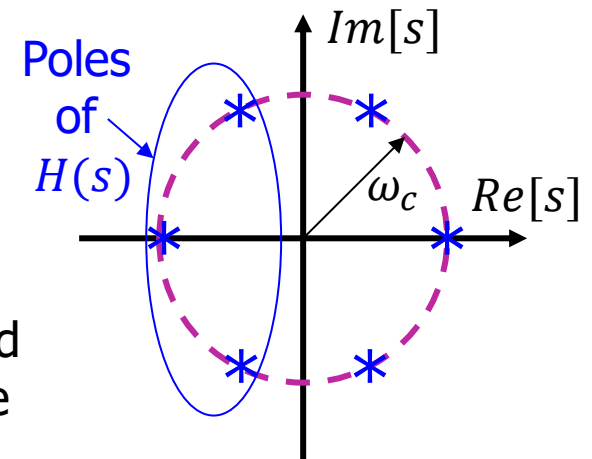
$$\rightarrow f = f_c(-1)^{1/2n} = f_c(e^{j(2k+1)\pi})^{1/2n} = f_c \exp\left(\frac{j(2k+1)\pi}{2n}\right), \quad \text{for } k = 0, 1, 2, \dots, 2n-1.$$

Or, in the s-domain (using $s = j\omega = j2\pi f$) the potential poles of the transfer function are at

$$s = j2\pi f_c \exp\left(\frac{j(2k+1)\pi}{2n}\right) = \omega_c \exp\left(\frac{j(2k+n+1)\pi}{2n}\right),$$

for $k = 0, 1, 2, \dots, 2n-1$.

Of these $2n$ potential pole locations, n belong to $H(s)$ and n belong to $H^*(s)$. In order for the filter to be stable, the poles in the LHP are selected to belong to $H(s)$.



Butterworth LPF Construction



To construct an n^{th} order Butterworth LPF with a cutoff (3dB) frequency of f_c , follow these steps:

Step 1 – Determine the order, n and cutoff frequency, ω_c , needed (more on that later).

Step 2 – Form the n pole locations,

$$p_k = \omega_c \exp\left(\frac{j(2k+n+1)\pi}{2n}\right),$$

for $k = 0, 1, 2, \dots, n-1$.

Step 3 – Form the transfer function

$$H(s) = \prod_{k=0}^{n-1} \left(\frac{-p_k}{s - p_k} \right) = \prod_{k=0}^{n-1} \left(\frac{1}{1 - \frac{s}{p_k}} \right).$$

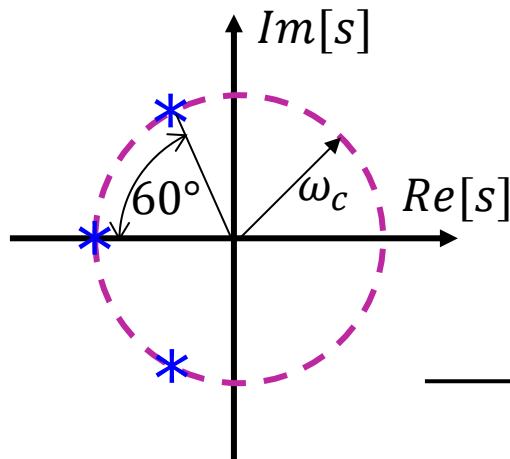
Note: Complex poles will occur in conjugate pairs.

Example 15.3: Form the transfer function for a 2nd Order Butterworth LPF

Butterworth LPF Construction



Example 15.4: Form the transfer function for a 3rd Order Butterworth LPF



$$H_1(s) = \frac{\omega_c}{s + \omega_c}$$

$$H_2(s) = \frac{\omega_c^2}{s^2 + s\omega_c + \omega_c^2}$$

Butterworth LPF Implementation



In general, an n^{th} order Butterworth LPF can be implemented as a series of first and second order real filters.

- ✓ For even n , use $n/2$ second order filters,
- ✓ For odd n , use $(n - 1)/2$ second order filters + one first order filter.

The n^{th} order Butterworth LPF transfer function can be decomposed as (for $\omega_c = 1$):

$$H_n(s) = \begin{cases} \prod_{k=1}^{n/2} \frac{1}{s^2 + 2s\cos(\theta_k) + 1}, & \text{for } n \text{ even,} \\ \frac{1}{s+1} \cdot \prod_{k=1}^{(n-1)/2} \frac{1}{s^2 + 2s\cos(\theta_k) + 1}, & \text{for } n \text{ odd.} \end{cases}$$

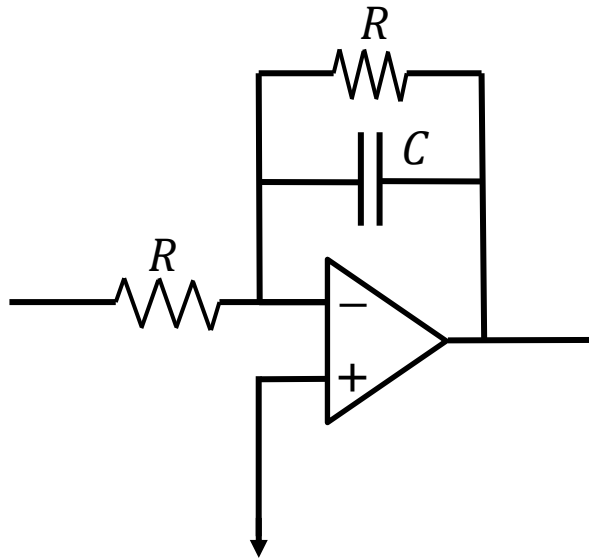
n	2	3	4	5	6	7	8	9
θ	45°	60°	22.5° 67.5°	36° 72°	15° 45° 75°	25.71° 51.43° 77.14°	11.25° 33.75° 56.25° 78.75°	20° 40° 60° 80°

For other values of ω_c , just replace s with $\frac{s}{\omega_c}$ in the above expressions.

Filter Implementation



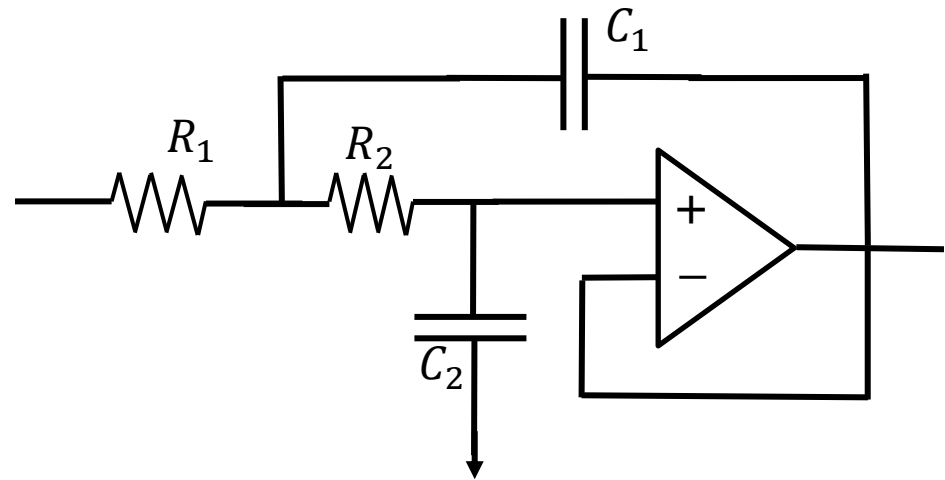
The first order filter can be implemented as an RC filter and the second order stages as RLC filters, but it is often preferable to build these using op-amp circuits.



First Order Stage

$$H(s) = \frac{-1}{1 + sRC}$$

$$RC \rightarrow \frac{1}{\omega_c}$$



Second Order Stage (Sallen-Key)

$$H(s) = \frac{1}{R_1 C_1 R_2 C_2 s^2 + (R_1 + R_2) C_2 s + 1}$$

$$R_1 C_1 R_2 C_2 \rightarrow \frac{1}{\omega_c^2} \quad (R_1 + R_2) C_2 \rightarrow \frac{\cos(\theta_k)}{\omega_c}$$